



Is it time to bring back chaos in cities?

Paolo Testolini on rethinking urban grids in the era of AI.

In the world of urban design, the grid has long been king. The planning grid, often referred to as a grid street plan or gridiron plan, has its origins in ancient city planning but became notably more systematic during the 18th and 19th centuries.

One of the earliest known uses of a grid layout dates to around 4500 BCE in ancient cities like Mohenjo-Daro in the Indus Valley Civilization. These cities had grid-like streets, which were among the first to use systematic urban planning.

The ancient Greeks, particularly the city of Miletus in the 5th century BCE, adopted grid planning for urban layouts. This method of organizing cities with intersecting streets in right angles became more common, providing an efficient way to manage traffic and growth.

Roman cities often followed the grid system, with Rome itself adopting the approach as the empire expanded. Roman planners designed grid streets around

south) to **grow systematically** but also to be able to **defend** their cities from invasions.

The orthogonal grid, with its neat intersections and predictable blocks, has dominated city planning from New York City's streets to Dubai's modern sprawl. It's been the backbone of efficiency—until now. With the rise of driverless cars and smart mobility, the question must be asked: Is the grid outdated? Could it be time to ditch the rigid streets and bring back the chaotic, meandering world of organic city planning?

The myth of the perfect grid

For centuries, city planners have believed that order must come first. The orthogonal grid, with its right angles and predictable intersections, promised a world of efficiency: easy navigation, straightforward expansion, and uniform infrastructure. **For those in charge of urbanization, the grid was a symbol of modernity—an emblem of control over chaos.** But when you think about it, is this really how we want cities to evolve?

The grid was a response to a specific set of challenges: increased urbanization and heavy vehicle traffic. But with autonomous cars and advanced smart mobility on the horizon, do these same principles apply? We're entering an age where driverless vehicles can communicate with each other, reducing traffic and the need for infrastructure-heavy road systems. Shouldn't our urban designs reflect this shift?

The organic grid

In the ancient Middle East, there were also grid-planned cities – Babylon, for example, had a more complex urban layout that included elements of a grid system.



Ancient plan of the city of Babylon (600 – 540 BCE).

Source: Ludwig Maximilian University of Munich; Royal Ontario Museum (ROM)

The city's layout, while not as systematically gridded as some later cities, featured large straight streets that divided the city into manageable blocks. This layout supported the flow of trade and communication and is thought to have influenced urban planning in the region.

But in general, ancient Middle Eastern cities often adapted their layouts to their environments (e.g., based on river systems, natural terrain, or the requirements of defense), elements of grid planning were present and influenced later city designs in the region.

Organic versus planned grid



In the context of historical Middle Eastern cities, the organic grid generally worked better in terms of adapting to the region's geography, climate, and social practices. Many cities grew organically around central hubs like markets, mosques, and plazas, and their street systems evolved in response to the landscape. In this context, organic streets allowed for more intimate urban experiences, offered shade, and preserved the city's connection to its surroundings.

However, in more recent times, especially in modern urban planning, the orthogonal grid has been increasingly used in Middle Eastern cities for its practicality in expansion, transportation, and organization. Cities like Abu Dhabi and Dubai in the UAE have adopted grid systems (with some organic elements), enabling rapid growth and modern infrastructure.

Ultimately, both systems have their place, and a hybrid approach is sometimes used: maintaining organic street patterns in historic districts or central areas, while employing orthogonal grids in newer developments to support growth and infrastructure.

In many Middle Eastern cities, an organic grid pattern has distinct advantages, particularly in its adaptability to geography. Many of these cities have been built around natural features such as rivers, hills, or desert landscapes, which make an organic layout more suitable. This kind of pattern can follow the topography and natural routes, making navigation more intuitive for people familiar with the landscape.

Additionally, cultural and historical factors have played a significant role in shaping the organic growth of these cities. During the Islamic Golden Age, which spanned from the 7th to the 13th centuries, traditional Middle Eastern cities grew organically, with streets developing gradually around religious, commercial, or residential hubs, such as mosques or markets. This organic development allowed for flexibility and facilitated the expansion of neighborhoods and trading areas without rigid urban planning.

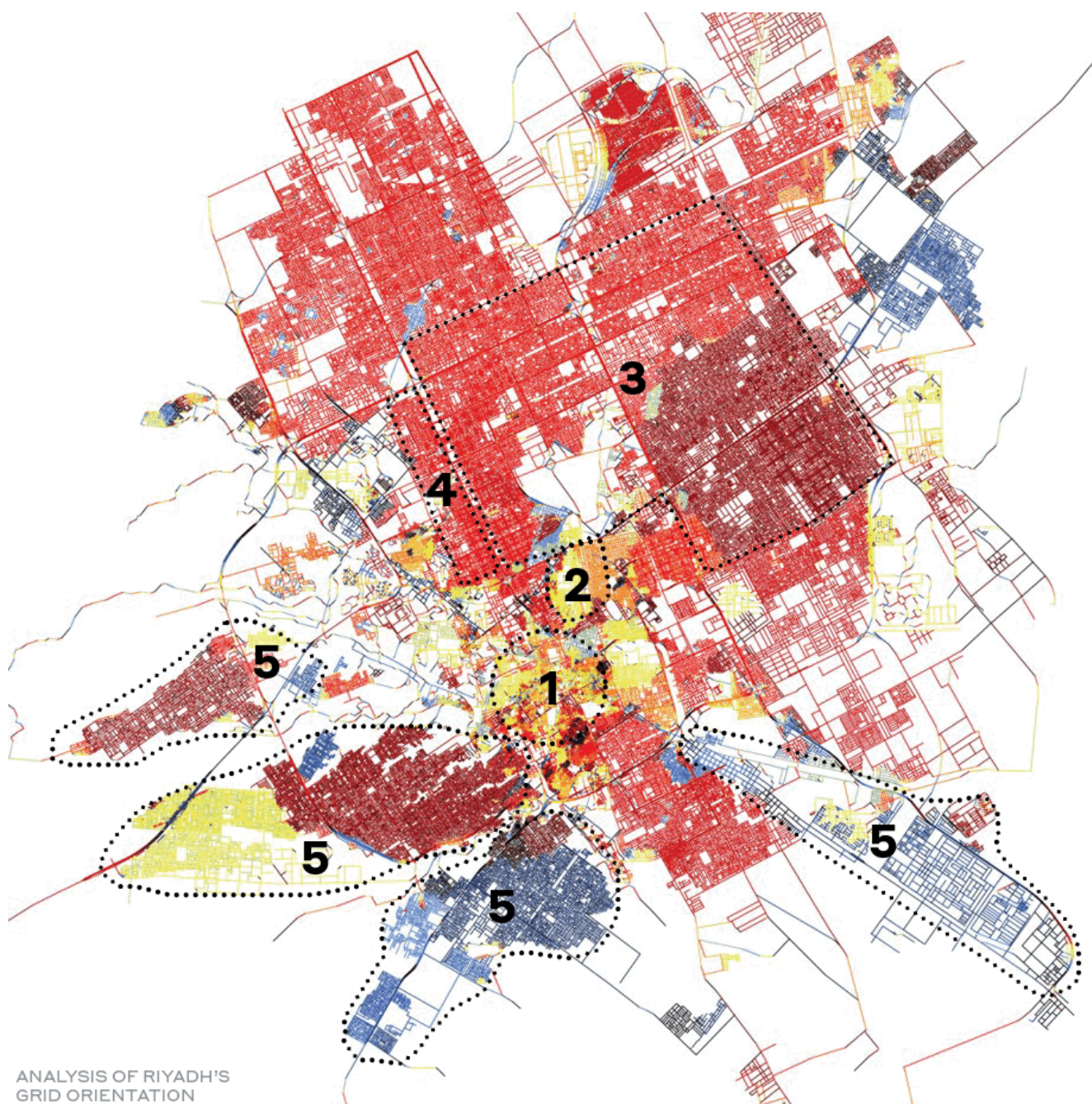
In hot desert climates, such as those found in the Middle East, organic street patterns have the added benefit of shading and cooling. The narrower, winding streets typical of an organic layout limit direct sunlight, helping to cool the area and create more comfortable walking conditions during the heat of the day. However, despite these advantages, organic layouts also have several disadvantages. As cities expand, the lack of a structured layout can lead to a chaotic street plan, making navigation more difficult, especially for newcomers. Furthermore, implementing and maintaining infrastructure, such as roads, water systems, and sanitation, becomes more challenging in organically grown cities.

angles to each other, forming a predictable, organized grid. This deliberate, planned approach simplifies navigation and makes it easier to expand and organize a city. One of the key advantages of the orthogonal grid system is its efficiency and order. It divides the land into regular blocks, which helps with both navigation and planning, particularly for newcomers. The clear, linear layout of an orthogonal grid also facilitates the development of infrastructure like transportation networks, utilities, and sewage systems.

Another advantage of the orthogonal grid is its capacity for expansion and growth. As cities develop, the grid system can easily accommodate new subdivisions and urban areas in a logical and predictable way. The regularity of the grid also promotes the efficient organization of different zones, such as residential, commercial, and industrial areas, which can in turn foster trade and better land use.

However, this system is not without its drawbacks. In the hot, arid climates of the Middle East, the wide streets that typically accompany an orthogonal grid may not be as effective at providing shade, leaving them more exposed to the sun. This can result in a less comfortable environment compared to the narrow, shaded streets of an organic grid. Additionally, the rigidity of the orthogonal grid can be a disadvantage when adapting to natural features or historical buildings. If imposed on a landscape that doesn't suit it—such as one with rivers, mountains, or uneven terrain—it may lead to inefficient land use.

Case Study: Riyadh's evolution from organic to orthogonal



Riyadh has undergone a remarkable evolution throughout the 20th century, with its urban grid layout reflecting distinct phases of its growth. The city's road grid and spatial orientation patterns reveal this layered history, which has transformed the city from a modest settlement into a sprawling, modern metropolis with its challenges and opportunities.

The historic core of Riyadh, where the city first began, consists of an organic layout that is characteristic of early settlements across the Middle East **(1)**. This organic growth reflects the necessity for adaptability in the desert environment and the economic drivers of the time. These paths connected key areas like the historic Diriyah, the heart of Riyadh's early settlement, to the city's markets, palaces, and places of worship.

In the 1930s, when Riyadh became the capital of Saudi Arabia, the city expanded beyond its ancient walls, with urban planning shifting towards more structured, orthogonal grids. These grids spread to the north and west, giving rise to areas like

The 1970s oil boom brought about another significant shift in Riyadh's growth trajectory. The city's rapid expansion was driven by newfound wealth and the influx of workers, leading to the creation of large-scale infrastructure projects. During this time, King Fahad Road emerged as a vital commercial corridor, stretching northward from the historic center and connecting key districts, further cementing the orthogonal grid's dominance in Riyadh's urban design **(3)**. This period of growth was crucial in shaping the city's modern identity, with major developments, highways, and commercial centers springing up along these key arterial roads **(4)**.

In recent decades, Riyadh's development has spread outward, beyond the confines of the ring road. The city's urban fabric now extends along major routes, but it also encounters disruptions, such as large plots like King Salman Park (a former military airbase), industrial zones to the east, and natural wadi formations that create barriers on the western edge of the city. These elements continue to influence Riyadh's growth, adding both complexity and opportunity to its urban development.

At the city's periphery, beyond the ring road, urban development takes on a more dispersed, tendril-like form **(5)**. This more organic spread signals a shift away from rigid planning models and reflects Riyadh's current phase, where the integration of nature, infrastructure, and urban life is a key design consideration. The city's evolving road grid, from organic to orthogonal and now to more expansive, is a direct reflection of the forces shaping Riyadh today—ambitious development, rapid growth, and the search for balance between modernity and tradition.

The rise of the autonomous city

Imagine a world where cars no longer need to follow fixed lanes or sit in traffic jams. With driverless cars, smart streets, and real-time data, traffic bottlenecks could be a thing of the past. And if this is the case, why should we cling to the idea that every street needs to be aligned to a grid?

Autonomous vehicles don't need wide roads with sharp turns or rigid intersections. Self-driving cars can optimize their routes, communicating seamlessly to avoid congestion and navigate efficiently through less predictable road networks. Organic street layouts—with winding, meandering paths that evolve with the environment—could suddenly become far more efficient than the stale orthogonal grid. No longer will we be stuck in the straitjacket of blocky streets; fluidity could reign in future urban designs.

What Does the Street of the Future Look Like?

in a world where driverless cars and smart mobility redefine how we move through cities; the street could become a place where community spirit thrives. Smaller, slower streets could encourage interaction and engagement, making them spaces for socializing, sharing, and gathering. With cars no longer dominating the roadways, we could reclaim these spaces for pedestrians, cyclists, and public transit.

Instead of vast parking lots and wide, underused roads, we could create green corridors lined with trees, public squares, and art installations that encourage walking, biking, and socializing. These streets would be designed for people first, not just cars, with plenty of space for community interaction. Imagine public markets, outdoor cafes, and open spaces nestled in urban districts where people live, work, and play—all supported by the seamless mobility of driverless cars and shared transportation.

Walkability and happiness in organic grids

One of the most significant advantages of organic grids is their potential to foster walkability—a key ingredient in enhancing quality of life and promoting happiness in cities. Walkable cities are proven to improve physical health, reduce stress, and promote social interaction. Organic layouts, with their narrower streets and more intimate spaces, naturally encourage walking, as pedestrians can move through neighborhoods more fluidly, without the monotony of long, straight roads.

In contrast to grid-heavy cities, which often prioritize the needs of cars over pedestrians, organic grids place people at the heart of the urban environment. These street designs foster connectedness and engagement, offering opportunities for spontaneous interaction. Public spaces—such as parks, plazas, and community gardens—become places where people can come together, fostering a sense of belonging and well-being. And the more people walk, the more they benefit, not just physically but emotionally, too. This human-centric design leads to happier, healthier cities.

How organic grids fight climate change

Organic grids are also better suited to the demands of climate change and global warming. As cities grapple with rising temperatures, extreme weather events, and increased urban heat islands, designing for a changing climate becomes crucial. Organic grids—by virtue of their meandering streets—can naturally create shaded areas, reduce wind tunnel effects, and promote cooler microclimates in dense urban environments.

In contrast, orthogonal grids, with their wide streets and open, sun-exposed intersections, are not as effective at providing shade or reducing the urban heat island effect. Organic layouts often provide more opportunities for green

cool the environment and contribute to overall urban resilience.

Moreover, organic grids can make cities more adaptable to the changing climate. The flexibility inherent in these systems allows urban spaces to evolve in response to environmental needs over time. Whether it's expanding green areas, introducing water management solutions, or redesigning streets to mitigate flooding, organic grids allow for a dynamic approach to addressing climate challenges.

Could organic city plans actually be more efficient?

Let's face it: grids are boring. They're predictable. Monotonous. And, quite frankly, they've become outdated. Sure, they work in some places, but in others, they feel like a desperate attempt at control in a world that craves complexity. But what if that complexity could be managed—smartly, efficiently, and without the chaos?



With driverless cars, the unpredictable organic street patterns from centuries past could make a stunning comeback. Think of cities like Venice or Marrakech—cities with curved alleys, irregular intersections, and natural flow. They may not have been designed with autonomous technology in mind, but in a world where cars communicate in real time and don't need parking spaces every few feet, these organic layouts suddenly become functional again. No more gridlock. No more wasted space for parking. Just cities that grow organically—like nature intended.

designs. But as technology evolves, we now have a unique opportunity to rethink the urban landscape. Autonomous cars can work seamlessly in less structured environments, and smart city technology will manage traffic, power, and services far more effectively than old-school planning ever could.

Smart infrastructure and smaller roads

What if we didn't need the wide roads of old? Driverless cars could navigate narrower paths, optimizing routes based on traffic conditions, weather, and passenger needs. Instead of devoting massive amounts of land to parking lots and expansive roads, organic grids would allow for more intimate, walkable spaces with narrower, more charming streets that still serve all the needs of modern life.

Imagine a city where the buildings and streets are designed with human connection in mind—where technology enables us to be closer to what really matters. With driverless cars reducing the need for oversized streets, we could reclaim space for public parks, green zones, and community hubs. The result? A city that feels alive, not just efficient.



The end of the grid?

In the face of autonomous vehicles and smart mobility, the orthogonal grid's days may be numbered. As technology continues to advance, it becomes increasingly clear that city planning no longer has to be dictated by straight lines and right angles. Instead, complexity, flow, and organic growth could shape the future of urban spaces.

So, let's ask ourselves the tough question: Do we really need the grid anymore? Or is it time to embrace the chaos, trust in technology, and let cities evolve as

The grid may have served us well for centuries, but with the tools of tomorrow, it's time to ask if it's time to tear it up and make room for something better.

The Era of AI: Reinventing cities for the 21st century

With the era of AI now upon us, the possibilities for reimagining cities are more exciting than ever. What if we could combine the best of both worlds—the organic grids of medieval cities with the efficiency and technology of the 21st century? AI could guide urban planners in creating adaptable, human-centric cities that respond dynamically to environmental, social, and technological changes. Just as medieval cities thrived on their organic growth and human scale, we can now use AI to ensure these organic grids are as efficient as possible, optimizing everything from traffic flow to energy consumption. Cities could evolve naturally, like living organisms, while still benefiting from the smart technologies and sustainability solutions needed to thrive in the modern world.

The future of urban planning is here—the organic grid, reimagined for the AI era—where the chaos of the past meets the efficiency of the future.



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Based in Dubai studio, Paolo oversees the ERA-co global urban design and masterplan studios. With more than 20 years of experience leading Architecture,

master planning combines evidence-based methodology with insight into the way people relate to spaces.

Paolo has directed master planning, mixed-use, retail and residential projects around the world—from Italy, Greece, Mexico, China and the UK. Currently, he focuses on large-scale projects in the Middle East—including the Dubai Land Retail District, the Palm Jumeirah Masterplan and the Expo 2020 Souks masterplan in Dubai. Paolo is a skilled public speaker. He regularly collaborates with the Council of Tall Buildings and Urban Habitats, the Welsh School of Architecture and Italy's Roma Tre University by teaching courses and speaking at design conferences.

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